Google Search Analysis Project

Introduction:

The **Google Search Analysis Project** focuses on analyzing search trends to understand what people are searching for and how interest in different topics changes over time. With the rapid growth of the internet, analyzing search behavior helps businesses, marketers, and analysts identify trending topics, user interests, and seasonal patterns.

In this project, search trend data was collected using **Google Trends** through the Python library **Pytrends**. The analysis was performed using powerful data analytics tools such **as Python, Pandas, Matplotlib, and Plotly** to process, analyze, and visualize search trend data.

Objective:

The main objective of this project is to identify:

1) First, write a code where, by changing just the keyword, we can search for multiple things.

2) Top 15 countries where the keywords are searched the most, and also create visual representation of it.

3) A world map needs to be plotted showing the countries that search the keyword the most.

4) We need to extract the time-wise interest of the keyword — how it trended in different years.

5) Compare related keywords and plot the graph.

<u>Analysis:</u>

1. **First, write a code where, by changing just the keyword, we can search for multiple things.**

   Ans- I have setup **Pytrends Library** and Defined the Keyword. The Keyword is most demanding that is **Data Science**.

2. **Top 15 countries where the keywords are searched the most, and also create visual representation of it**.

   Ans- The chart shows **which countries search the most for the term "Data Science."** The data is usually obtained from **Google Trends** using the **Pytrends** library in **Python**.

   🥇 **1. India – Interest ≈ 100**

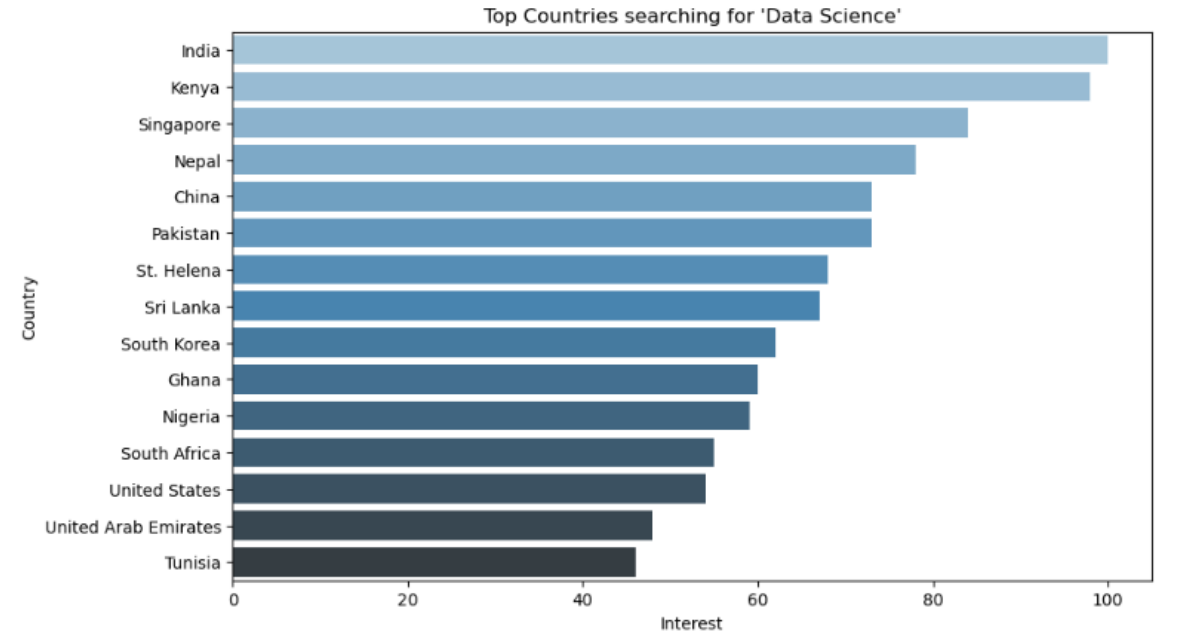
- Highest search interest.
- Indicates **very high demand for data science learning and jobs**.
- Many students and professionals are searching for data science courses and careers.

🥈 **2. Kenya – Interest ≈ 98**

- Strong interest in data science.
- Shows growth in **tech and analytics education**.

🥉 **3. Singapore – Interest ≈ 85**

- Major **technology and analytics hub** in Asia. High demand for **AI and data professionals**.



3. **A world map needs to be plotted showing the countries that search the keyword the most.**

   Ans- This is a **Choropleth Map (Geographical Heat Map)**.

- Each country is colored based on **search interest level**.
- Darker colors indicate **higher interest**.
- Lighter colors indicate **lower interest**.

  This type of visualization is commonly used in **data analytics and business intelligence dashboards**.

3⃣ Color Scale (Legend on the Right)

The color bar shows the **interest score from about 45 to 100**.

Interpretation:

- **100 → Highest search interest**
- **80–90 → Very high interest**
- **60–80 → Moderate interest**
- **Below 60 → Lower interest**

Important:
These values represent **relative popularity**, not the actual number of searches.

4⃣ Countries with Highest Interest

🇮🇳 India – Score ≈ 100

- Darkest color on the map.
- Indicates the **highest search interest in "Data Science."**
- Large number of students and professionals searching for **data science courses and jobs**.

🇰🇪 Kenya – High interest

- Also shown with a dark shade.
- Indicates growing interest in **technology and analytics education**.

5⃣ Countries with Moderate Interest

Countries such as:

- China
- Nepal
- Pakistan
- Sri Lanka

show **moderate to high interest** in the topic.

These regions are experiencing:

- Growth in **AI and machine learning**
- Increasing **tech education**
- Rising demand for **data-related jobs**

6⃣ Countries with Lower Relative Interest

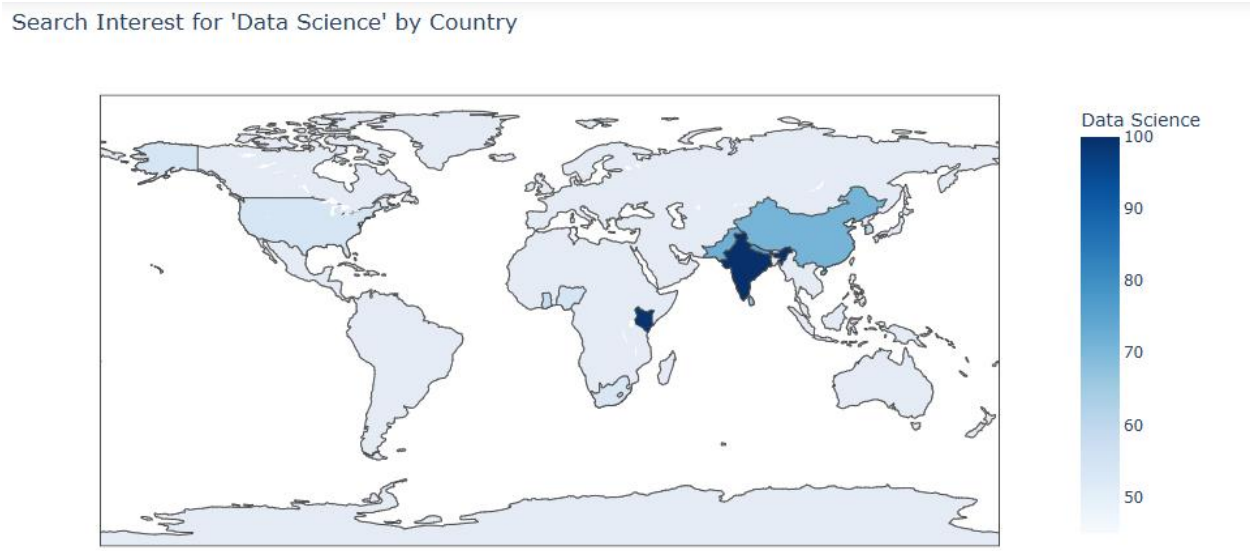
Countries like:

- United States
- South Africa
- Brazil

appear lighter on the map.

This **does NOT mean low usage**.
Instead, it means:

- People search many other topics.
- The **relative share of "Data Science" searches is smaller** compared to other topics.



4. **We need to extract the time-wise interest of the keyword — how it trended in different years.**

Ans- **1⃣ Search Interest Over Time – "Data Science"**
What the Chart Shows

This line graph shows **how the popularity of the keyword "Data Science" changed over time**.

- **X-axis:** Date (from 2025 to 2026) and **Y-axis:** Interest level (0–100)

Meaning of Interest Score

- **100 → Maximum search popularity**
- **50 → Moderate interest**
- **Below 40 → Lower interest**

These scores are **relative values**, not actual search numbers.

Trend Analysis

Early Period (March – June 2025)

Interest remains **moderate between 33–40**.

This suggests:

- Stable but not very high search demand.

Growth Phase (July – September 2025)

Interest increases rapidly:

- July: around **45–50**
- August–September: **80+**

Possible reasons:

- Increased interest in **data science courses**
- Academic semesters starting
- Growing industry demand

Fluctuation Phase (October – December 2025)

Interest fluctuates between:

- **60 – 85**

This indicates **consistent global interest in Data Science**.

Peak Period (Early 2026)

The graph reaches its **highest value close to 100**.

This suggests:

- Major spike in search activity
- Possibly related to **job trends, courses, or AI discussions**

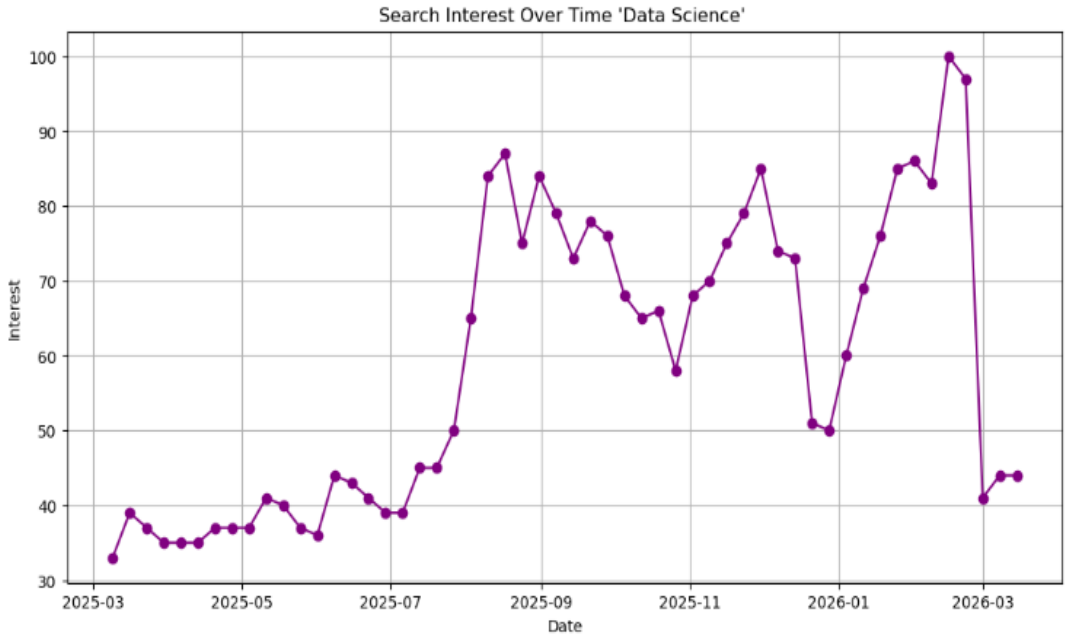
Sudden Drop

At the end of the timeline, the interest drops sharply to around **40–45**.

Possible reasons:

- Short-term spike earlier
- Seasonal search behavior

Key Insight:
Search interest for **Data Science** has **generally increased over time**, showing strong and growing demand for this field.



5. **Compare related keywords and plot the graph.**

Ans- This graph compares **four technology keywords**:

- Data Science
- Machine Learning
- Cloud Computing
- Artificial Intelligence

All keywords are commonly used in modern technology and analytics.

X-axis
Date (2025–2026)
Y-axis
Interest score (0–100)

Keyword Trend Analysis

Data Science

Moderate but steady interest.

- Range: **25 – 55**
- Shows **consistent popularity**

Used in:

- analytics
- business intelligence
- predictive modeling

Machine Learning

Highest overall interest.

Peak near **100**, making it the **most searched topic** among the four.

Reasons:

- AI revolution
- automation
- predictive algorithms

Machine Learning is a **core part of modern AI systems**.

Cloud Computing

Lowest search interest among the four.

Range: **5 – 30**

However, it remains important because it powers:

- cloud infrastructure
- big data platforms
- scalable applications

Artificial Intelligence

Second highest trend after Machine Learning.

Range: **20 – 75**

Interest increases significantly during **AI boom periods**.

AI includes:

- robotics
- natural language processing
- computer vision

Important Observations

1. Machine Learning dominates searches

It reaches the **highest interest score**, indicating strong global curiosity and adoption.

2. AI and Data Science grow together

Their trends rise during similar periods because they are closely related fields.

3. Cloud Computing grows slowly

Even though search interest is lower, it remains essential for **data storage and large-scale computing**.

Business / Industry Insight

This type of analysis helps:

Companies

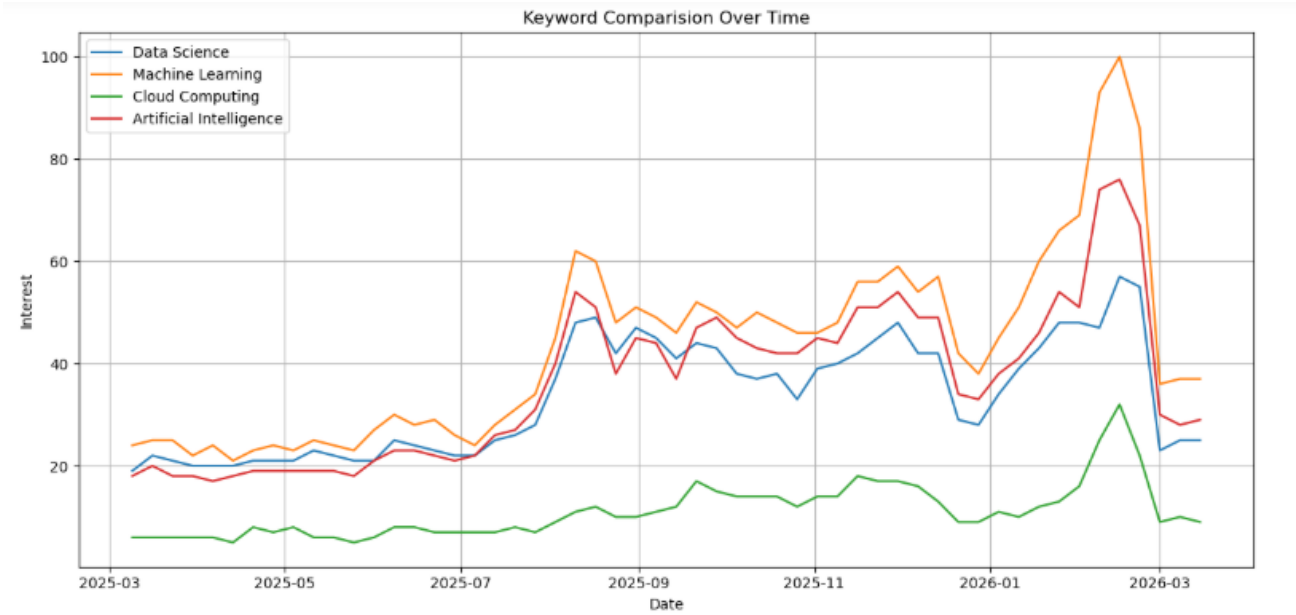
Understand **technology trends**.

Education Platforms

Create courses based on **popular skills**.

Job Seekers

Identify **high-demand technology skills**.



Conclusion:

By analyzing these trends, the project demonstrates how **search data can provide valuable insights for digital marketing, business decision-making, and market research**. The visualizations and statistical analysis help transform raw search data into meaningful insights that can guide companies in understanding consumer interests and predicting future trends.